



Adversarial Examples in Deployment

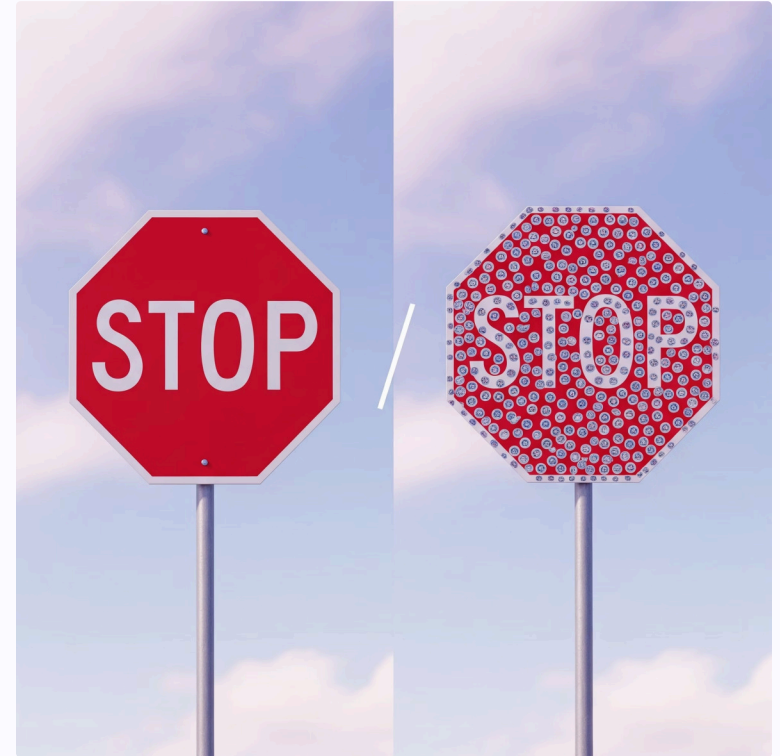
Small, crafted perturbations that cause catastrophic model errors represent one of the most studied vulnerabilities in AI security today, threatening vision, NLP, speech, and autonomous systems in production.

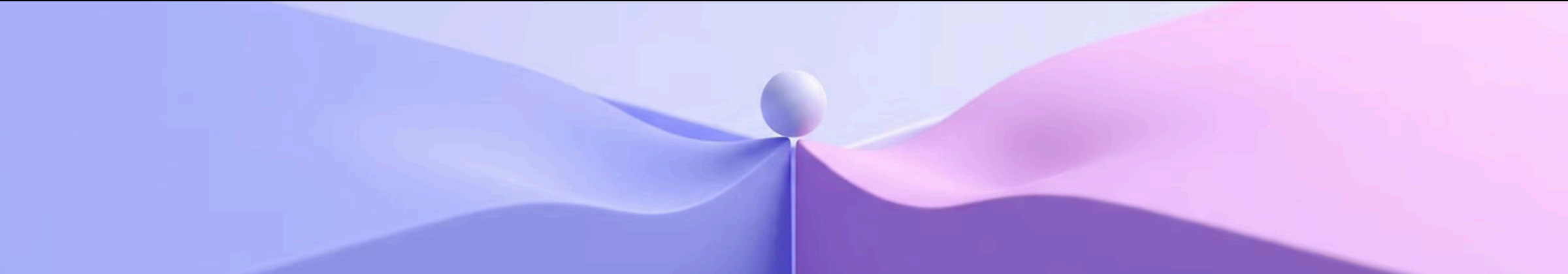
What Are Adversarial Examples?

Adversarial examples are inputs modified with **imperceptible noise** that cause models to predict incorrectly with **high confidence**.

While humans see normal input, the model is completely fooled.

Example: A stop sign with carefully placed stickers is misclassified as a speed-limit sign by a computer vision model.





Why They Work

Fragile Decision Boundaries

Deep learning models rely on decision boundaries that can be exploited with minimal changes to input data.

High-Dimensional Space

Small perturbations in high-dimensional feature spaces can easily cross classification boundaries.

Optimization

Attackers systematically optimize perturbations to maximize model error while remaining imperceptible.

Overfit Models

Models that lack robustness are particularly susceptible to these precisely crafted inputs.

Attack Types

Evasion Attacks

Adversarial noise added during inference to bypass detection

- Occurs at test-time
- No modification to the model
- Most common type

Poisoning Attacks

Backdoors planted during model training

- Manipulates training data
- Creates specific vulnerabilities
- Hard to detect after deployment

Physical Attacks

Perturbations in real-world objects

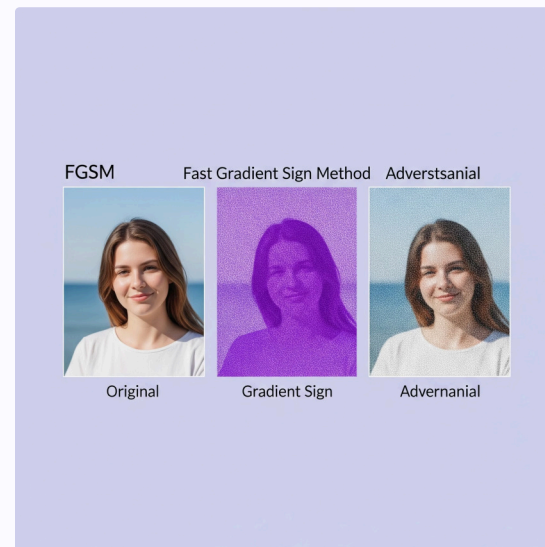
- Stickers on objects
- Specialized glasses frames
- Adversarial audio patterns

Example: FGSM Attack

Fast Gradient Sign Method

$$x' = x + \epsilon \cdot \text{sign}(\nabla_x L(\theta, x, y))$$

- One-step perturbation along gradient sign
- Easy & fast to compute
- Demonstrates fundamental model fragility
- Created by Goodfellow et al. in 2014



The attack adds noise in the direction that maximizes loss, scaled by epsilon.

Example Code: FGSM in PyTorch

```
import torch

def fgsm_attack(image, epsilon, data_grad):
    # Get the sign of the gradient
    sign_data_grad = data_grad.sign()

    # Create the perturbed image
    perturbed_image = image + epsilon * sign_data_grad

    # Ensure valid pixel range [0,1]
    return torch.clamp(perturbed_image, 0, 1)
```

✓ **Expected result:** A slightly modified image that appears identical to humans but causes the model to misclassify with high confidence.

📘 This attack is surprisingly effective against many state-of-the-art models despite its simplicity, revealing fundamental vulnerabilities in neural networks.

White-Box vs Black-Box Attacks

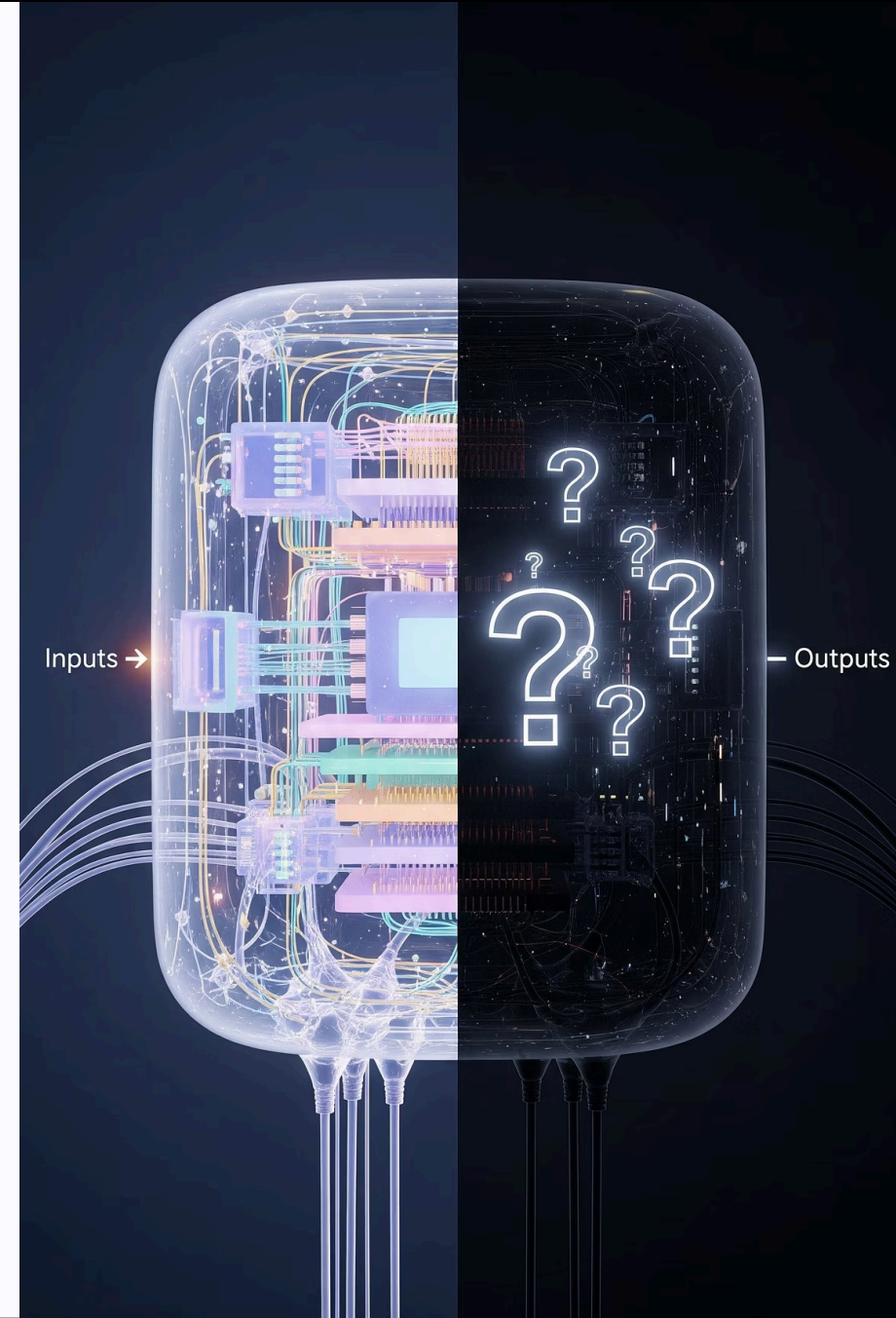
White-Box Attacks

- Attacker has complete knowledge of model architecture
- Access to weights, gradients, and parameters
- Examples: FGSM, PGD, C&W attacks
- Most powerful attack scenario

Black-Box Attacks

- Attacker only has query access to model API
- Utilizes transferability of adversarial examples
- Often trains surrogate model to craft attacks
- More realistic in deployed systems

Both approaches remain effective if robust defenses are not implemented.



Domains at Risk



Autonomous Vehicles

Road signs perturbed with adversarial stickers can cause misclassification, creating potential traffic accidents and safety hazards.



Finance

Small tweaks to transaction features can bypass fraud detection systems, enabling financial crimes to go undetected.



Healthcare

Adversarial X-rays or medical images can fool diagnostic AI systems, potentially leading to missed diagnoses or improper treatments.



NLP

Adding synonyms or character modifications can flip sentiment classification, content moderation, or chatbot responses.



Physical Adversarial Examples

Real-World Manifestations

- Printed stickers on physical objects confuse vision models
- Specially designed eyeglass frames defeat facial recognition
- Audio perturbations trigger "Hey Siri" or other voice commands

Key Threat

These attacks extend beyond digital manipulation, creating **practical threats in physical environments** where AI systems are increasingly deployed.

Physical-world robustness is harder to guarantee than digital robustness.

Defenses Against Adversarial Examples

1

Adversarial Training

Retrain model with adversarial samples included in the training set to develop immunity to similar attacks.

Effective but computationally expensive and may not generalize to novel attacks.

2

Defensive Distillation

Train a second model on softened outputs of the first model to smooth decision boundaries and reduce sensitivity.

Creates smoother gradients that are harder to exploit.

3

Input Preprocessing

Apply denoising, compression, or transformation to detect or remove adversarial perturbations before classification.

Can be circumvented if attacker is aware of defense.

4

Certified Defenses

Mathematical guarantees that certain perturbations cannot change model output beyond specified bounds.

Provides provable robustness but typically for limited types of perturbations.

Example: Adversarial Training in PyTorch

```
# Generate adversarial examples during training
for x, y in train_loader:
    x.requires_grad = True
    output = model(x)
    loss = criterion(output, y)
    model.zero_grad()
    loss.backward()
    x_adv = fgsm_attack(x, 0.1, x.grad.data)

    # Train on both clean + adversarial examples
    optimizer.zero_grad()
    output_adv = model(x_adv)
    loss_adv = criterion(output_adv, y)
    (loss + loss_adv).backward()
    optimizer.step()
```

✓ **Expected outcome:** Model becomes more robust to FGSM-like attacks by learning to classify correctly even in the presence of adversarial perturbations.

This approach expands the training distribution to include adversarial examples, improving model resilience.

Infrastructure Implications



1 Attack Surface

Deployed APIs represent a potential vulnerability point that attackers can probe and exploit.

2 Monitoring Requirements

Systems need real-time detection of unusual query patterns that may indicate adversarial probing.

3 Life-Critical Applications

Latency-sensitive applications like self-driving vehicles and healthcare diagnostics face **potentially fatal consequences** from adversarial attacks.

Robust infrastructure design must integrate adversarial detection at multiple levels.

Research Case Studies

2017: Physical Adversarial Examples

Researchers demonstrated that stop sign adversarial stickers could reliably mislead vision models in autonomous vehicles, proving physical-world attacks were practical.

Eykholt et al., "Robust Physical-World Attacks on Deep Learning Models"

1

2

2018: Audio Attacks

Hidden commands embedded in audio signals were shown to activate voice assistants without human detection, raising concerns about security of smart home devices.

Carlini & Wagner, "Audio Adversarial Examples"

3

2020+: NLP Adversarial Attacks

Word substitution attacks demonstrated vulnerabilities in BERT-based models, showing that synonyms could dramatically change classification outcomes while preserving meaning.

Jin et al., "Is BERT Really Robust?"

The Impact of Small Perturbations

"Small noise, big mistake."

These visual examples demonstrate how imperceptible changes to inputs can completely undermine AI system reliability, highlighting the gap between human and machine perception.

What humans ignore, models amplify.



Summary

Key Takeaways

- Adversarial examples exploit fundamental model fragility
- Attacks span white-box, black-box, and physical domains
- Defenses exist but the security arms race continues
- Robust deployment requires both monitoring and architectural defenses

Next Steps for ML Engineers

1 Test & Monitor

Implement adversarial testing in pre-deployment validation

2 Defense in Depth

Layer multiple defense mechanisms rather than relying on one approach

3 Risk Assessment

Evaluate potential impact of adversarial examples in your specific domain

Adversarial robustness is not optional for production ML systems.